

When Do LLM Code Agents Commit to a Strategy? A Counterfactual Measure of Strategic Commitment via Conditional Outcome Entropy

Ramtin Sirjani

Western University

Working draft — Introduction & Methods (pre-experiment); June 11, 2026

Abstract

Test-time diversity helps multi-step language-model agents explore alternative solutions, but in a trajectory of dozens of steps it is unclear *where* injecting diversity is worth the compute. We hypothesize that an agent progressively *commits*: as its trajectory unfolds the distribution over final solutions narrows (mode collapse increasing with accumulated context). We introduce a causal, non-circular measure of strategic commitment: the *conditional semantic entropy of the final strategy given the trajectory prefix*, $H(S \mid \tau_{:k})$, estimated by fixing the first k steps, resampling M independent continuations to completion, and measuring the semantic entropy of their final outcomes. This is the agent-trajectory, entropy-generalized analogue of the *committor* from transition-path theory—the probability a path reaches one basin before another—but for a *learned* generator, which forces a distinction the physical committor never had: the prefix f^* where the model’s *own* resampling collapses (it stops exploring) can precede the prefix $f^\dagger$ where the outcome is causally *locked* (no intervention can change it). The object we actually track is the per-step *leverage*—the mutual information between a step and the final outcome—whose elementary bound $\ell_k \leq \min(\text{step branchiness, residual outcome entropy})$ explains why diversity pays only at a few, possibly scattered, steps rather than a contiguous window; $f^* \leq f^\dagger$ summarize it, ordered rigorously through reachable *support* (not an entropy inequality), and we align trajectories by the fraction of outcome entropy resolved rather than by wall-clock step. Because a single model conflates an early-committing *problem* with a mode-collapsed *model*, we estimate the curve across a panel of models, yielding an *outcome landscape* that separates the two and locates where solution diversity lives—within a model versus across models. This document is a methods-and-design draft: it states the measure, the interventional estimator, and the strategy-similarity instrument the estimator relies on, and *pre-registers* the experimental protocol prior to running it. No outcome results are reported.

1 Introduction

Problem setting. We study the *autonomous* agent setting, not the interactive one. The agent is not engaged in a multi-turn back-and-forth with a user; instead it receives a single fixed input—a code repository together with a natural-language issue describing a bug—and then runs unattended. It interleaves private reasoning with tool calls (reading code, editing files, executing the test suite) over a long trajectory, and emits exactly one output: a code patch, scored against held-out tests [3, 2]. There is no human in the loop after the initial input, and no intermediate answers are returned; the entire trajectory is the model’s own, and its only product is the final patch. Formally, a sample is a function of the input and the model’s sampling randomness alone, $\tau = \text{Agent}(\text{repo, issue}; \omega)$, terminating in a patch $S = \text{patch}(\tau)$. This is what makes “where to inject diversity” a nontrivial question: unlike a single-shot prompt, the stochasticity is spread across dozens of autonomous steps, and unlike a dialogue, there is no user turn at which to intervene.

Test-time diversity. Multi-step LM agents of this kind solve realistic software-engineering tasks by interleaving reasoning and tool use over long trajectories [1, 2, 3]. A recurring way to improve them at inference time is *diversity*: sample several trajectories and keep the best, or branch the search to cover distinct solution paths [4, 5, 6, 7]. Diversity is only useful, however, when the samples are genuinely different; modern instruction-tuned models exhibit substantial mode collapse, producing only a handful of functionally distinct outputs even at high temperature [8].

The *where* problem. In a single-shot setting one injects diversity once, at the answer. A multi-step agent is different: diversity can be injected at *any* step of a long trajectory, and branching at every step is exponentially expensive. Tree- and MCTS-style agents branch either uniformly or under a learned value heuristic [5, 6], but neither asks the prior question: *where in a trajectory is diversity actually worth the compute?* Spending branches in the wrong place wastes test-time compute; spending none at the right place forgoes the gains diversity can offer.

The commitment intuition. We hypothesize that an agent’s trajectory *progressively narrows* its outcome distribution: early steps leave many solutions open, and each additional fixed step tends to concentrate probability on fewer of them, so the conditional outcome entropy falls as the prefix grows—mode collapse increasing with accumulated context. This is the agentic analogue of an effect observed for single-shot generation, where output entropy can fall and the model “locks in” toward the end of a sequence—though the pattern is itself *model-dependent*, decreasing for some models and not others [15], which already foreshadows our cross-model analysis (§3.5)—and it is consonant with documented diversity collapse in aligned models [14, 8]. We are deliberately *not* claiming that once an approach is chosen the remainder is mere execution: even with the overall approach fixed, its implementation can still vary, so the narrowing is gradual and *granularity-dependent*—the *approach* (which location, what mechanism) may lock in well before the precise *behavior* does (§3.4). Two cautions keep this honest. First, “where the outcome is still *open*” is not the same as “where diversity is *decided*.” At the very first step the outcome is maximally open, yet the model’s opening action is often near-deterministic given the repository—it reads the obviously-relevant file—so forcing a different first action has no purchase on the outcome even though nothing is yet fixed. What licenses an injection at a step is that step’s *leverage* on the final outcome, a quantity distinct from how open the outcome remains. Second, neither narrowing nor leverage need be monotone: alignment and what was just read make some mid-trajectory steps far branchier than their neighbours, and a step that inspects code can even *raise* outcome uncertainty by revealing several viable repairs. The realistic picture is therefore not a single wide-to-narrow sweep but an uneven profile in which a few *scattered* steps carry most of the commitment; forced diversity pays at those steps—branchy enough that plain resampling misses the alternatives, yet still consequential enough that the choice reaches the outcome—wherever they fall. We formalize “leverage” in §3.2 and make this the primary object; whether the high-leverage steps cluster into a band or are dispersed, and where, is the empirical question this measure is built to answer.

Why the obvious measure is circular. The tempting way to locate commitment is observational: distill the agent’s *stated* strategy at each step and ask when it stops changing. This is both circular—it measures drift toward the realized final state, which any trajectory exhibits as it ends—and, more subtly, it conflates levels of a hierarchical strategy. An agent’s *overall approach* (which function to change, by what mechanism) may fix early while its *substrategy* (locate → implement → test → debug) legitimately evolves throughout. A measure that watches the whole evolving description will therefore always see change.

Our measure: counterfactual commitment. We define commitment *causally*. The strategy is committed at step k if, fixing the prefix $\tau_{:k}$ and resampling the continuation, the final strategy no longer varies. This is

precisely the conditional entropy of the outcome given the prefix, $H(S \mid \tau_{:k})$. We estimate it by intervention (Figure 1): fork the trajectory at prefix length k , run M independent continuations to completion, cluster their final outcomes by meaning, and compute the discrete semantic entropy. The resulting *commitment curve* $\hat{H}(S \mid \tau_{:k})$ versus k is high where the strategy is still open—forking there yields diverse outcomes—and collapses to zero once the strategy is committed (Figure 2). It references no “final” anything: it is the divergence of *counterfactual futures*, a do-style intervention on the prefix. Formally this is an *interventional, entropy-generalized committor*: transition-path theory defines the committor $q(x)$ —the probability a trajectory from microstate x reaches the product basin before the reactant—as the ideal one-dimensional reaction coordinate, zero in one basin and one in the other [17, 18]. Our $\hat{H}(S \mid \tau_{:k})$ replaces the two-basin probability with the entropy over K executed-outcome classes and the physical dynamics with a sampled *learned* policy; that last substitution is exactly what splits the committor’s single collapse into two thresholds (§3.2), because a learned generator can stop exploring before the outcome is dynamically fixed.

One experiment, two uses. The same intervention that *locates* commitment also *prescribes* where to inject diversity, unifying a descriptive finding about agent behavior with an actionable test-time-compute policy. But the prescription is not simply “branch where $\hat{H}(S \mid \tau_{:k})$ is high.” Where $\hat{H}(S \mid \tau_{:k})$ is high the model already diversifies under plain resampling, so forcing diversity buys little; where $\hat{H}(S \mid \tau_{:k})$ has collapsed, resampling is exhausted yet the outcome may still be *reachable* by a forced perturbation. Branching pays at the high-leverage steps—branchy enough that resampling misses the alternatives, still consequential enough to reach the outcome—which lie between f^* and $f^\dagger$ but need not form a contiguous band; the same forking apparatus locates them by contrasting plain continuations against forced-diverse ones (§3.2).

Relation to test-time-compute allocation. A growing line allocates inference compute by *local* uncertainty. Entropy-gated branching forks at high-entropy *reasoning* steps and prunes the expansions with a verifier [11]; confidence-aware scaling for web agents triggers extra compute when an action’s *vote-distribution* entropy or margin crosses a threshold [13]; and learned policies decide *when to plan* from a trained signal [12]. These signals are cheap and online but *correlational*: high local uncertainty need not change the final outcome, and low local uncertainty can still precede divergent outcomes. We instead measure the *interventional* conditional entropy of the *final* outcome given the prefix—the ground-truth quantity such gates approximate—not to build a cheaper gate but to map *where* outcome diversity actually lives in a trajectory, against which those gates can be calibrated. The observation that branching at uncertain junctures yields more diverse continuations [11] is, in our terms, the statement that those junctures fall in the open window before lock-in.

Contributions.

1. A causal, non-circular measure of strategic commitment in multi-step LM agents—the conditional semantic entropy of the final outcome given the trajectory prefix—together with an interventional estimator based on deterministic prefix replay and continuation resampling (§3.2–§3.3).
2. A *checkable, execution-grounded* meaning-equivalence relation for code-agent outcomes: because patches are programs we run, we cluster them by their *behavioral signature* (the test-outcome vector they induce), with an optional *structural* refinement (which files/functions/hunks change). Both are deterministic and require no learned text-similarity and no model judgment—a relation unavailable to text-only semantic-entropy work (§3.4).
3. A *cross-model* reading of the measure: because one model conflates genuine commitment with mode collapse, pooling outcomes over a model panel estimates a problem’s *outcome landscape*, against which a model’s coverage—and the split between *outcome* diversity and *path* diversity—becomes interpretable (§3.5).

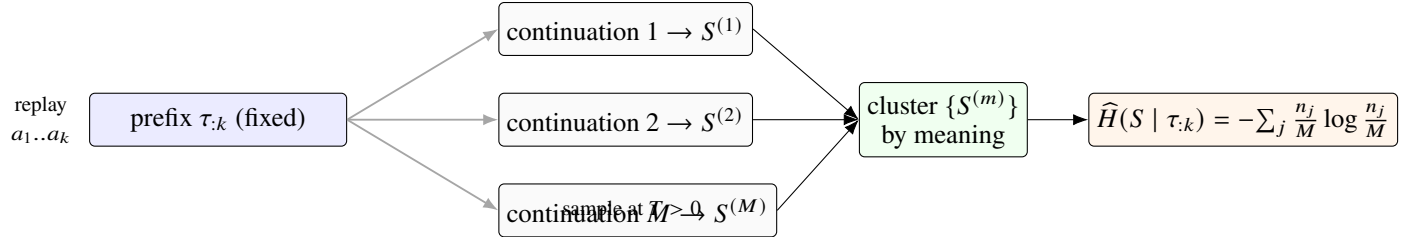


Figure 1: Interventional estimator of $H(S | \tau_{:k})$. The first k steps are fixed (reconstructed by deterministic replay), then M continuations are sampled to completion. Their final strategies $S^{(m)}$ are clustered by meaning and the discrete semantic entropy is computed. Sweeping k traces the commitment curve.

4. A pre-registered experimental protocol to estimate commitment curves and the outcome landscape across SWE-bench problems, locating the diversity window and testing the early-commitment hypothesis (§3.6). *Results are not included in this draft.*

2 Related work

Commitment as a collapsing curve is an old idea. Across fields, the uncertainty of a sequential process’s outcome falls as the process unfolds, after which the result is effectively determined. Chemical physics makes this precise with the *committor* of transition-path theory: $q(x)$, the probability a trajectory from microstate x reaches the product basin before the reactant, runs from 0 to 1 along a reactive path; its $q = \frac{1}{2}$ iso-surface is the *transition state*; and it is regarded as the ideal one-dimensional reaction coordinate [17]. The information thermodynamics of the transition-path ensemble further ties entropy production to how much the dynamics reveal about a trajectory’s reactivity [18]. The same collapsing shape appears as sports win-probability curves that flatten as a game progresses [19] and as the empirical predictability of LLM reinforcement-learning trajectories from early states [20]. We adopt this lineage explicitly: $\hat{H}(S | \tau_{:k})$ is an entropy-generalized committor for agent trajectories. What it adds is a feature absent when—as in physics—the sampler and the dynamics are the same object: a *learned* generator can stop exploring before the outcome is dynamically fixed, splitting the single collapse into the pair $(f^*, f^\dagger)$.

Irreversibility and option value in RL. The $f^\dagger$ side—an outcome becoming unreachable—is the subject of reversibility-aware RL, which learns which action sequences cannot be undone [21], and of the *power/empowerment* literature, where keeping many futures reachable is instrumentally valuable [22] and is now estimated for language-model agents directly [23]. These quantify the reachability of *states*; we instead track the entropy of the final *executed outcome*, and we pair its causal-lock threshold with the generator’s collapse threshold—a separation the reachability view does not make.

Diversity and forking in language models. Closest to our mechanism, Bigelow et al. [24] resample single-turn generations token-by-token and use Bayesian change-point detection to locate *forking tokens* where the final-answer distribution shifts, summarized by a *survival function* $S(T)$. We differ on four axes that matter for the agent setting: their analysis is single-turn and token-level, ours is over a multi-step agent’s trajectory prefixes; their outcomes are one-hot answers, LLM-extracted text, or embeddings, whereas ours are *executed* behavioral signatures (the test-outcome vector a patch induces), giving a non-circular, non-learned meaning relation; they measure one curve under forced resampling, we separate the generator-collapse (f^*) and outcome-lock ($f^\dagger$) thresholds; and we turn the result into a test-time-compute policy (inject in $[f^*, f^\dagger]$). A parallel training-time line finds that a sparse set of high-entropy “forking”/“critical” tokens decides reasoning

direction [25, 26] and trains models to branch at them [27]; these corroborate our premise—decision-bearing positions are sparse and the diverse-but-correct forks sit deep in the trajectory, predicting a non-trivial f^* —but they operate on the training loss, remain single-turn, and never measure *where* branching pays. Finally, the inference-time uncertainty gates for agents discussed above [11, 13, 12] are complementary: they approximate, online and locally, the interventional quantity we measure offline.

3 Methods

3.1 Setup and what we mean by “strategy”

We study a ReAct-style agent [1] that solves a task by emitting a sequence of steps $\tau = (a_1, o_1, \dots, a_T, o_T)$, where each a_t contains reasoning and a single shell action and o_t is the resulting observation; the episode terminates when the agent submits a patch. Our primary model is Qwen3-Coder-30B served locally; for the cross-model landscape analysis (§3.5) we add a panel of open code models. Tasks are drawn from SWE-bench Verified [3] (SymPy), with one isolated container per trajectory.

The outcome of a trajectory is its **final patch**, and the entropy in §3.2 is taken over *meaning classes* of patches. The crucial point is that, because our outputs are executable programs, “same meaning” is not a free-text judgment but a *checkable* one: two patches mean the same thing if they make the repository *behave* the same way. We define the meaning map as a hierarchy from coarse to fine (§3.4): (i) **resolution**—resolved vs. not (one bit); (ii) **behavioral signature**—the vector of per-test outcomes a patch induces, optionally augmented by hashed differential-test outputs; and (iii) **structural**—which files, functions, and hunks the patch changes. All three are deterministic and require no learned text-similarity. Behavioral equivalence (ii) is the principled backbone: two textually different patches that flip the same tests are the same outcome (renamings, reordered independent edits, caller- vs. callee-side fixes), and two near-identical patches that differ on a test are different outcomes—distinctions that surface-level diffing gets backwards. This relation is unavailable to text-only semantic-entropy work; here we can run the code. Structural equivalence (iii) is a finer, deterministic refinement used as a secondary lens—it separates two genuinely different *approaches* that happen to behave identically. In all cases the patch—not the long, scaffold-dominated, order-sensitive trajectory—is the unit of meaning.

3.2 Conditional outcome entropy and commitment

Let the agent define a stochastic policy π at sampling temperature T , so the final strategy S (the patch) is a random variable. Fix a prefix $\tau_{:k} = (a_1, o_1, \dots, a_k, o_k)$ of length k . We define the conditional semantic entropy of the outcome given the prefix,

$$H(S \mid \tau_{:k}) = - \sum_{c \in C} p(c \mid \tau_{:k}) \log p(c \mid \tau_{:k}), \quad p(c \mid \tau_{:k}) = \Pr_{\text{cont} \sim \pi(\cdot \mid \tau_{:k})} [\text{meaning}(S) = c], \quad (1)$$

where C is the set of *meaning classes* of final patches and the meaning map is the behavioral (primary) / structural hierarchy of §3.4; we evaluate $\hat{H}(S \mid \tau_{:k})$ at each granularity. Following discrete semantic entropy [9, 10], the quantity is defined over meaning classes rather than surface forms and requires no token probabilities. Read along the reference trajectory, $\hat{H}(S \mid \tau_{:k})$ is the *cumulative* diversity still reachable from step k by plain resampling—a statement about the outcome, not yet about any individual step’s role in deciding it.

What licenses an injection: per-step leverage. The cumulative curve is not yet the object we want, because injecting diversity at step k changes the outcome only insofar as step k actually *decides* it. The quantity that measures this—our *primary* object—is the conditional mutual information between the step’s action A_k and

the final outcome, given the prefix, $\ell_k = I(A_k; S \mid \tau_{:k-1})$. By the elementary bound that mutual information cannot exceed either marginal entropy [16],

$$\ell_k \leq \min (H(A_k \mid \tau_{:k-1}), H(S \mid \tau_{:k-1})). \quad (2)$$

Leverage is appreciable only where the step is itself *branchy* ($H(A_k \mid \tau_{:k-1})$ large—the model has genuinely different actions available) *and* the outcome is still *open* ($H(S \mid \tau_{:k-1})$ large—there is something left to decide). This is the formal content of the balance between conditioning and uncertainty: leverage vanishes at both ends of a trajectory for *opposite* reasons—at deterministic “boilerplate” steps (read the obvious file, run the suite) the first term is ≈ 0 , so the step decides nothing however open the outcome; once the outcome is fixed the second term is 0, so even a branchy step is inconsequential. In particular, injecting at the very first step is typically wasteful not because diversity is cheap there but because the opening action is forced by the repository ($H(A_1 \mid \tau_{:0}) \approx 0$) even though the outcome is maximally open. The profile $\{\ell_k\}$ is therefore generally uneven and multi-peaked, a few scattered steps carrying most of the commitment; we do *not* assume a single wide-to-narrow sweep. Averaged over a step’s realizations ℓ_k equals the expected drop in outcome entropy (chain rule of mutual information [16]), but along a *single* reference the realized increment $H(S \mid \tau_{:k-1}) - H(S \mid \tau_{:k})$ can be *negative*—a step that inspects code may reveal that several repairs are viable—so it is only an expectation-level proxy for ℓ_k .

Two summary thresholds, stated honestly in support. Two scalars bracket where injection can help. They are cleanest in terms of the *support* of the outcome distribution (the classes with positive probability), for which a rigorous ordering holds; entropy alone does not give one. Let $\text{supp}_\pi(k)$ be the outcome classes reachable by plain resampling from $\tau_{:k}$, and $R(k)$ those reachable by *any* admissible intervention on the next step followed by resampling. Plain resampling is one such intervention, so $\text{supp}_\pi(k) \subseteq R(k)$ at every k . Define

$$f^* = \sup\{k/T : |\text{supp}_\pi(k)| > 1\}, \quad f^\dagger = \sup\{k/T : |R(k)| > 1\}, \quad (3)$$

the *last* prefix fraction at which plain resampling, respectively any intervention, still yields more than one outcome (the last upcrossing, not the first dip, since the curves are non-monotone). The inclusion $\text{supp}_\pi \subseteq R$ gives $f^* \leq f^\dagger$ *rigorously*: if no intervention can move the outcome, plain resampling certainly cannot. We deliberately avoid the stronger-looking claim that the forced entropy dominates the natural one pointwise—forcing one particular action can *lower* outcome entropy—so the inequality we rely on is the support inclusion, with entropy used only as a graded read-out. The *injectable set* $\{k : |\text{supp}_\pi(k)| = 1 < |R(k)|\}$ —resampling collapsed but an intervention still reaches alternatives—lies in $[f^*, f^\dagger)$ but need *not* be a contiguous interval. The claim $f^* < f^\dagger$ is exactly that the model commits (its resampling support collapses) before the problem locks (its reachable support collapses); $f^\dagger$ is reported relative to the forcing operator actually used—a lower bound on true reachability (§3.6).

A reaction coordinate: aligning by resolved entropy. Indexing by prefix fraction k/T is lossy: the same decision can occur at step 3 of one trajectory and step 7 of another, so averaging by k/T smears distinct decisions together. Transition-path theory meets the same problem and indexes not by time but by the committor, the reaction coordinate measuring progress through the transition [17]. Its analogue here is the *fraction of outcome entropy resolved*,

$$\rho(k) = 1 - \frac{H(S \mid \tau_{:k})}{H(S \mid \tau_{:0})} \in [0, 1] \quad (H(S \mid \tau_{:0}) > 0), \quad (4)$$

0 when nothing is decided, 1 at lock-in. We report the leverage profile and the thresholds against ρ as well as k/T , so commitment events align across trajectories by *how far the outcome has been decided* rather than by wall-clock step. (ρ is monotone only in expectation and estimated with noise; we use it as a coordinate, never as a quantity to optimize.)

Estimator. From a prefix $\tau_{:k}$ we draw M continuations $\text{cont}^{(1)}, \dots, \text{cont}^{(M)} \sim \pi(\cdot \mid \tau_{:k})$, cluster their final patches into classes $\{C_j\}$ of sizes n_j , and form the plug-in discrete entropy

$$\widehat{H}(S \mid \tau_{:k}) = - \sum_j \frac{n_j}{M} \log \frac{n_j}{M}, \quad \widehat{H}(S \mid \tau_{:k})^{\text{norm}} = \widehat{H}(S \mid \tau_{:k}) / \log M. \quad (5)$$

Evaluating this at successive prefixes along the reference traces the natural curve $H(S \mid \tau_{:k})$ and its support $\text{supp}_\pi(k)$; repeating with a forced maximally-diverse first action traces the reachable surrogate $R(k)$. The per-step leverage ℓ_k is read two ways: cheaply, as the curve increment $\Delta_k = \widehat{H}(S \mid \tau_{:k-1}) - \widehat{H}(S \mid \tau_{:k})$, an expectation-level proxy that can go negative; and, on a cost-limited subset of steps, by a *direct nested* estimate of the mutual information itself—draw R alternative actions $a_k^{(1)}, \dots, a_k^{(R)} \sim \pi(\cdot \mid \tau_{:k-1})$, continue each M times to outcome distributions $\hat{p}_r$, and form the Jensen–Shannon-type information $\widehat{\ell}_k = H(\frac{1}{R} \sum_r \hat{p}_r) - \frac{1}{R} \sum_r H(\hat{p}_r)$. Plug-in entropies are biased low at small M ; we apply the Miller–Madow correction and report $\log M$ as the ceiling. The estimator is **non-circular** (it references only counterfactual continuations, never the realized final) and **interventional** (fixing $\tau_{:k}$ and resampling is a $\text{do}(\cdot)$ on the prefix). It is monotone **only in expectation**: $\mathbb{E}[H(S \mid \tau_{:k})]$ is non-increasing in k because conditioning cannot raise average entropy [16], but the single-reference curve we measure can rise locally—which is why the thresholds are defined by sustained collapse and the leverage profile, not a first crossing, is the object of record.

A second axis: path entropy. The curve $\widehat{H}(S \mid \tau_{:k})$ measures the diversity of *where* a trajectory ends. Orthogonal to it is the diversity of *how* it reaches a given end—the path-conditional entropy $H(\text{path} \mid S=c)$, which we summarize by the dispersion (pairwise structural similarity and length) of the continuations that land in a fixed class c . A class reached by many dissimilar paths is robustly “canalized-to”; one reached by a single path is brittle. Outcome entropy and path entropy are independent—a problem may admit one outcome reached a hundred ways, or ten outcomes each reached one way—and a single scalar conflates them, so we report both (§3.5).

3.3 Reconstructing the prefix by deterministic replay

Sampling $\pi(\cdot \mid \tau_{:k})$ requires placing the agent in the exact state after k steps—crucially, the container filesystem written by $a_1, \dots, a_k$, not merely the textual history. We reconstruct this state by *deterministic replay*: we instantiate a fresh evaluation container and re-execute the recorded shell actions $a_1, \dots, a_k$ (deterministic file-reading/writing commands), which reproduces the on-disk state, and we set the agent’s message history to $\tau_{:k}$. Replay avoids storing a container image per step and yields the exact prefix state; continuations then proceed by ordinary sampling at temperature T . A single reference trajectory per problem (greedy decoding, $T = 0$) defines the prefixes to be reconstructed.

3.4 Meaning equivalence: behavioral and structural signatures

The entropy in Eq. (5) is taken over *meaning classes*, so it depends entirely on the relation that decides when two final patches mean the same thing. Because our outputs are executable, we take this relation to be **behavioral** as the principled primary: we run the patched repository against the instance’s tests and assign each patch its *behavioral signature*—the vector of per-test outcomes over the `FAIL_TO_PASS` and `PASS_TO_PASS` tests, optionally augmented by hashed outputs of additional differential inputs. Two patches share a class iff they share a signature. This collapses behaviorally-equivalent patches that surface diffing over-counts (renamings, reordered independent edits, caller- vs. callee-side fixes) and separates near-identical patches that flip a test, and it is obtained essentially for free: the evaluation harness already executes these tests. One caveat is inherent: behavioral equivalence is defined *relative to the test suite*—two patches agreeing on every

test may differ on untested inputs. For SWE-bench the curated suite is the task’s behavioral specification, so the relation is principled though coarse; differential or property testing sharpens it where needed.

A finer, *structural* refinement separates two genuinely different *approaches* that nonetheless behave identically—for instance a fix in the caller versus the callee that passes exactly the same tests. We obtain it deterministically from the patch itself: the set of files, functions, and hunks it touches (its structural signature), with two patches refined into the same class when they edit the same code locations in the same way. Crucially, this needs no *learned text-similarity*. The whole meaning hierarchy is therefore free of model judgment: the primary relation is *executed* (a test-outcome signature) and the refinement is *parsed* (a structural signature), so neither the generative model nor any auxiliary similarity model adjudicates what counts as “the same.” This is a deliberate departure from the text-similarity clustering of semantic-entropy work on free-form answers [9, 10], which is neither available nor necessary once the outputs are programs we can run and parse.

3.5 Beyond a single model: the outcome landscape

A single model’s commitment curve is confounded. A uniformly low $\widehat{H}(S \mid \tau_k)$ is consistent with two opposite situations: the problem genuinely admits one solution and the model commits to it immediately, or the problem admits many but the model is mode-collapsed onto one [8]. From one model these are indistinguishable—a flat low curve reads as early commitment but may be mere narrowness. The remedy is a *reference frame*. Let a panel of models $\{m_1, \dots, m_P\}$ each produce outcomes for a problem; merging their outcomes under the meaning relation of §3.4 yields the problem’s *outcome landscape* Λ , an estimate of the set of behaviorally distinct viable solutions. Against Λ the single-model quantities become interpretable: *commitment* corresponds to a small Λ (the problem really has few solutions), *collapse* to a large Λ of which the model covers only a corner. We summarize this with each model’s *coverage* $|C_m|/|\Lambda|$ and the cross-model class-overlap matrix, which separates *convergent* models (reaching the same classes) from *complementary* ones (reaching disjoint classes—the regime where ensembling helps).

Statistics logged. For each (model, problem) we record the per-step *leverage* profile $\{\ell_k\}$ (the curve increment Δ_k at every step, plus the direct nested estimate on a step subset), the natural and reachable outcome-support curves with the summary pair $(f^*, f^\dagger)$ and the injectable *set* between them, and the resolved-entropy coordinate $\rho(k)$; together with the *cost of commitment* (turns and tokens until the natural support first collapses to one class), the number of outcome classes, and, per class, its probability and intra-class path similarity (the path axis of §3.2). For each problem we record the landscape size $|\Lambda|$, per-model coverage, and the cross-model overlap matrix. All are computed from artifacts the runs already produce.

3.6 Pre-registered protocol

We fix the following before running, to avoid post-hoc tuning:

- **Reference & prefixes.** One greedy reference trajectory per problem. For the leverage profile we resample at *every* step where the trajectory is short and on a fixed stride otherwise (the stride is fixed in advance and reported), with $k=0$ the maximal-openness anchor; the cheap increment Δ_k is computed at all swept prefixes and the direct nested estimate $\widehat{\ell}_k$ on a pre-declared step subset.
- **Continuations.** M continuations per prefix at $T = 0.7$ (the model’s recommended sampling temperature), drawn twice per prefix—once plain (for $H(S \mid \tau_k)$ and the natural support supp_π , giving f^*) and once with a forced maximally-different next action (for the reachable support R , giving $f^\dagger$); the nested estimate additionally draws R actions per subset step. One container alive at a time.
- **Outcome & meaning.** Final patch; *primary* meaning relation = behavioral signature (per-test outcome vector); *secondary* = structural signature (files/functions/hunks changed); entropy by Eq. (5) at each

granularity, with Miller–Madow correction.

- **Models.** A primary model (Qwen3-Coder-30B) for the commitment curves; a panel of open code models for the outcome-landscape analysis (§3.5), served one at a time.
- **Problems.** The SWE-bench Verified SymPy set, scaling to a larger and more diverse array; all claims scoped to the set actually run.
- **Primary analysis.** The per-step leverage profile $\{\ell_k\}$ and the natural/reachable support curves, plotted against both k/T and the resolved-entropy coordinate ρ ; the joint distribution of $(f^*, f^\dagger)$ and the prevalence, size, and *contiguity* of the injectable set (is it a band or dispersed?); per-problem heterogeneity (early- vs. late-committing, Figure 2); and, across the panel, the outcome landscape $|\Lambda|$, per-model coverage, the path-entropy axis, and commitment-cost statistics (§3.5).

Hierarchy and ordering (secondary analysis). Because a strategy is an overall approach plus an ordered composition of substrategies, two trajectories can reach the same approach by different routes (reordered substeps) or diverge in substrategy while sharing an approach. The patch-level outcome answers the *overall-approach* question directly. To probe the substrategy layer we will additionally decompose each continuation into a sequence of sub-intents and compare sequences with an order-aware transport distance, reported separately; this layer is exploratory and does not affect the primary curve.

Anticipated threats to validity. (i) *Reference dependence:* f^* conditions on one reference prefix per problem; we will quantify sensitivity by repeating with sampled references on a subset. (ii) *Estimator bias and the leverage proxy:* small M biases plug-in entropy low (we report Miller–Madow-corrected values and $\log M$ as the ceiling); moreover the cheap increment Δ_k estimates per-step leverage only in expectation and can go negative along a single reference, so we treat it as a screen and confirm flagged steps with the direct nested estimate $\hat{\ell}_k$. (iii) *Support and coordinate sensitivity:* the support-based thresholds $f^*, f^\dagger$ depend on ε and M —a rare outcome class surfaces only with more samples and can push a threshold later—and ρ is computed from noisy entropy estimates; we report threshold sensitivity to (ε, M) and present results in both k/T and ρ . (iv) *Replay fidelity:* nondeterministic commands could perturb the reconstructed state; we restrict replay to the recorded deterministic actions and verify state equivalence on spot checks. (v) *Landscape under-coverage:* the outcome landscape Λ is a *lower bound* estimated from a finite model panel and finite sampling, so coverage figures are relative to it; a larger panel can only enlarge Λ . (vi) *Intervention dependence:* $f^\dagger$ and the injectable set are defined relative to a particular forced-diversity operator (a strategy-proposal injection), so they are a *lower bound* on true reachability—a stronger intervention can only push $f^\dagger$ later and widen the set; we report the operator explicitly and treat $f^\dagger$ as operator-relative, not absolute. (vii) *Scope:* results are scoped to the problem set and model panel actually run; the measure itself is model-agnostic.

Status

This draft presents the idea, the measure, and the protocol for review prior to any outcome experiments. The estimator, the prefix-replay harness, and the behavioral/structural meaning relations are implemented; the commitment-curve and outcome-landscape runs are withheld pending sign-off on the design.

References

- [1] S. Yao et al. ReAct: Synergizing Reasoning and Acting in Language Models. *ICLR*, 2023.
- [2] J. Yang et al. SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering. *NeurIPS*, 2024.

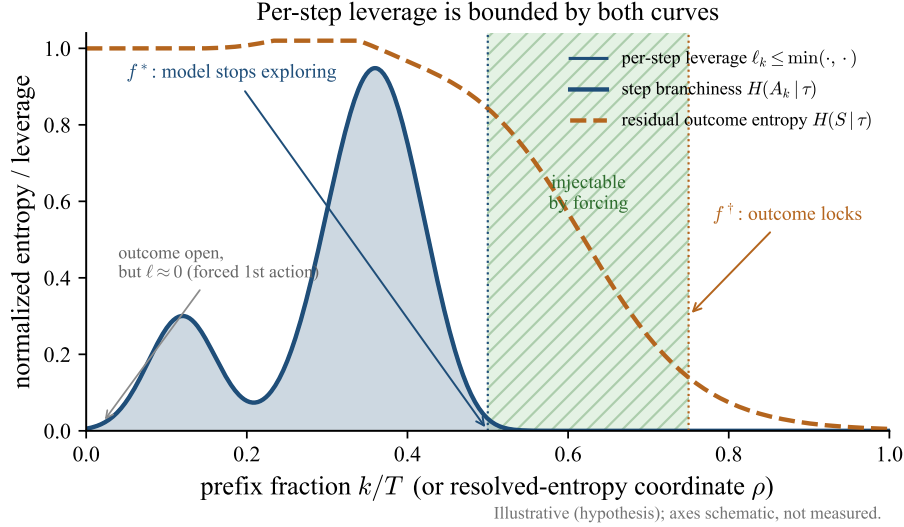


Figure 2: Hypothesis the experiment tests. Per-step leverage $\ell_k = I(A_k; S \mid \tau_{k-1})$ (filled) is bounded by *both* the step’s branchiness $H(A_k \mid \tau)$ (solid; is the model still exploring?) and the residual outcome entropy $H(S \mid \tau)$ (dashed; is anything left to decide?), so it is uneven and multi-peaked and vanishes at both ends for opposite reasons: the opening action is near-deterministic (leverage ≈ 0 though the outcome is wide open), and once the outcome is fixed even a branchy step is inconsequential. Branchiness collapses at f^* (the model stops exploring; natural support $\rightarrow 1$); residual entropy collapses later at $f^\dagger$ (the outcome locks; reachable support $\rightarrow 1$). The hatched region $[f^*, f^\dagger]$ is injectable by *forcing*—natural exploration is spent but the outcome is still reachable—and need not be contiguous. $f^* < f^\dagger$ is the model committing before the problem locks. *Schematic; not measured.*

- [3] C. Jimenez et al. SWE-bench: Can Language Models Resolve Real-World GitHub Issues? *ICLR*, 2024.
- [4] X. Wang et al. Self-Consistency Improves Chain of Thought Reasoning in Language Models. *ICLR*, 2023.
- [5] S. Yao et al. Tree of Thoughts: Deliberate Problem Solving with Large Language Models. *NeurIPS*, 2023.
- [6] A. Zhou et al. Language Agent Tree Search Unifies Reasoning, Acting, and Planning. *arXiv:2310.04406*, 2023.
- [7] C. Snell et al. Scaling LLM Test-Time Compute Optimally. . . *arXiv:2408.03314*, 2024.
- [8] Y. Zhang et al. NoveltyBench: Evaluating Language Models for Humanlike Diversity. *COLM*, 2025.
- [9] S. Farquhar et al. Detecting Hallucinations in Large Language Models Using Semantic Entropy. *Nature*, 630:625–630, 2024.
- [10] L. Kuhn et al. Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation. *ICLR*, 2023.
- [11] X. Li, E. Callanan, A. Ghassel, X. Zhu. Entropy-Gated Branching for Efficient Test-Time Reasoning. *arXiv:2503.21961*, 2025.
- [12] D. Paglieri, B. Cupiał, J. Cook, U. Piterbarg, J. Tuyls, E. Grefenstette, J. Foerster, J. Parker-Holder, T. Rocktäschel. Learning When to Plan: Efficiently Allocating Test-Time Compute for LLM Agents. *arXiv:2509.03581*, 2025.
- [13] N. Lee, L. E. Erdogan, C. J. John, S. Krishnapillai, M. W. Mahoney, K. Keutzer, A. Gholami. Agentic Test-Time Scaling for WebAgents. *arXiv:2602.12276*, 2026.
- [14] L. Yun, C. An, Z. Wang, L. Peng, J. Shang. The Price of Format: Diversity Collapse in Large Language Models. *arXiv:2505.18949*, 2025.

- [15] A. Adeseye, A. Adeseye, H. Tenhunen, J. Isoaho. Entropy and Attention Dynamics in Small Language Models: A Trace-Level Structural Analysis. *Intelligent Systems Conf. (IntelliSys)*, 2026.
- [16] T. M. Cover and J. A. Thomas. *Elements of Information Theory*. Wiley, 2nd edition, 2006.
- [17] W. E and E. Vanden-Eijnden. Transition-Path Theory and Path-Finding Algorithms. *Annual Review of Physical Chemistry*, 61:391–420, 2010.
- [18] M. D. Louwerse and D. A. Sivak. Information Thermodynamics of the Transition-Path Ensemble. *Physical Review Letters* (arXiv:2106.11511), 2022.
- [19] R. S. Brill, R. Yurko, A. J. Wyner. Exploring the Difficulty of Estimating Win Probability: A Simulation Study. *arXiv:2406.16171*, 2024.
- [20] Y. Cai, D. Cao, X. Xu, Z. Yao, Y. Huang, Z. Tan, B. Zhang, G. Sun, G. Liu, J. Fang. On Predictability of Reinforcement Learning Dynamics for Large Language Models. *arXiv:2510.00553*, 2025.
- [21] N. Grinsztajn, J. Ferret, O. Pietquin, P. Preux, M. Geist. There Is No Turning Back: A Self-Supervised Approach for Reversibility-Aware Reinforcement Learning. *NeurIPS*, 2021.
- [22] A. M. Turner, L. Smith, R. Shah, A. Critch, P. Tadepalli. Optimal Policies Tend to Seek Power. *NeurIPS* (arXiv:1912.01683), 2021.
- [23] J. Song, J. Gore, M. Kleiman-Weiner. Estimating the Empowerment of Language Model Agents. *arXiv:2509.22504*, 2025.
- [24] E. Bigelow, A. Holtzman, H. Tanaka, T. Ullman. Forking Paths in Neural Text Generation. *arXiv:2412.07961*, 2024.
- [25] S. Wang, L. Yu, C. Gao, C. Zheng, S. Liu, et al. Beyond the 80/20 Rule: High-Entropy Minority Tokens Drive Effective Reinforcement Learning for LLM Reasoning. *arXiv:2506.01939*, 2025.
- [26] Z. Lin, T. Liang, J. Xu, Q. Lin, X. Wang, R. Luo, C. Shi, S. Li, Y. Yang, Z. Tu. Critical Tokens Matter: Token-Level Contrastive Estimation Enhances LLM’s Reasoning Capability. *arXiv:2411.19943*, 2024.
- [27] S. Jia, X. Wang, S. P. Kasiviswanathan. Training Large Language Models To Reason In Parallel With Global Forking Tokens. *arXiv:2510.05132*, 2025.